
Beyond Components: From Singular Directions to Rank-2 Semantic Compasses

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Abstract

We ask what the atomic functional unit of a transformer attention head really is. Starting from the observation that individual OV singular directions in GPT-2 Small causally control the Gender-Pronoun (GP) task—a sparse set of five directions recovered by learnable masking drives GP accuracy from 84% to 32.67% under scalar steering, and to a full 100% flip at $\sigma \times 2.0$ —we show that these directions are not isolated 1D axes but live inside a coherent 2D geometry. For an attention head whose combined OV map has SVD $W_{OV} = U\Sigma V^\top$, pairs of columns (u_i, u_j) span *semantic compass planes* whose azimuthal angle codes a categorical distinction and whose radius codes its magnitude. An in-context rank-2 injection $\alpha\sigma(\cos\theta u_i + \sin\theta u_j)$ produces a sinusoidal logit-difference response whose amplitude grows linearly in α ($R^2 \geq 0.993$) and whose phase is stable to $\pm 3^\circ$ across $\alpha \in \{1, 3, 10\}$, in every one of four pretrained decoder-only models—GPT-2 Small, Phi-3 Mini, Gemma-2-2B, and Llama-3.2-3B. A blind causal scan over 864–4032 (L, H, u_i, u_j) candidates per model recovers the same heads without decoding, with head-level concentration significant at $p < 10^{-6}$ against a full-OV null, and the scan surfaces additional non-pronoun compasses in Phi-3 (month vs. hour, item vs. operation). The downstream GP-to-occupation transfer and the masculine-vs-feminine control asymmetry first observed at the 1D direction level both persist in the rank-2 setting and are partially but not fully closed by per-direction calibration. Together, this positions OV-SVD as an architecture-agnostic, weight-space primitive: singular directions carry causal content, and singular pairs carry a continuous θ -family of interventions.

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1. Introduction

Mechanistic interpretability aims to reverse-engineer the internal computations of neural language models. Two threads dominate: circuit analysis, which localizes behavior to *components*—specific attention heads or MLPs (Elhage et al., 2021; Wang et al., 2023; Conmy et al., 2023; Meng et al., 2022)—and linear feature discovery, which searches for low-dimensional causal *directions* in activation space (Park et al., 2024; Geiger et al., 2024; Marks & Tegmark, 2024; Engels et al., 2024; Cunningham et al., 2023; Templeton et al., 2024). A central question sits between them: what is the atomic functional unit? Whole heads are coarse; arbitrary residual-stream directions are not tied to any specific weight matrix. We argue the right unit is finer than a head but attached to the head’s own weights: the singular directions—and, more tightly, singular *pairs*—of its output-value map $W_{OV} = W_V W_O$.

This paper develops that claim in two stages.

Stage 1 (sub-component units). We first establish that individual OV singular directions are causal control axes. On GPT-2 Small and the Gender Pronoun (GP) task, a learnable-mask circuit discovery procedure applied to the SVD of each head’s OV and QK matrices isolates a sparse set of five gender-relevant directions. Continuous scalar steering along these directions produces monotonic dose-response curves: GP accuracy drops from 84.00% baseline to 32.67% under a single-sweep intervention and to 0.00% (a full 100% he→she flip) under $\sigma \times 2.0$ amplification. The control is *asymmetric*: masculine-coded directions steer the model more strongly than feminine-coded ones, and per-direction calibration of the feminine-only subset narrows but does not close this gap.

Stage 2 (rank-2 geometry). The directions recovered in Stage 1 are not isolated 1D axes. For head h at layer ℓ with SVD $W_{OV}^{(\ell,h)} = U\Sigma V^\top$, a *semantic compass* is a pair of columns (u_i, u_j) such that (i) vocabulary decoding of u_i and u_j through W_U yields a coherent angular ordering of category tokens, and (ii) residual-stream injection $\alpha\sigma(\cos\theta u_i + \sin\theta u_j)$ at the head’s own layer produces a logit-difference response $\text{LD}(\theta) \approx \mu + A \cos(\theta - \phi)$ whose amplitude scales linearly with α and whose phase is sta-

ble across α . The GP head GPT-2 L9H7 (direction SV_1 in Stage 1) and the GP head GPT-2 L10H9 (direction SV_0 in Stage 1) are both rediscovered as the azimuthal axes of genuine 2D compasses, (u_1, u_2) and (u_0, u_3) , satisfying the joint amplitude-linearity + phase-stability test. The same construction replicates in Phi-3 Mini (L24H10, L28H1), Gemma-2-2B (L21H4), and Llama-3.2-3B (L26H14), and surfaces two *non-pronoun* compasses in Phi-3: a temporal-unit dial (month vs. hour, L24H17) and an entity-type dial (item vs. operation, L24H28). A blind causal scan over 864–4032 candidate (L, H, u_i, u_j) planes per model, which never consults vocabulary decoding, recovers the same heads with head-level concentration significant at $p < 10^{-6}$ against a full-OV null. Finally, rotating the compass toward its feminine pole *and* using it without retraining on occupation prompts transfers the Stage-1 behavioral effect to the rank-2 setting, and the asymmetric-control phenomenon persists.

Contributions.

1. **Sub-component causal units (Section 3).** We recover a sparse set of OV singular directions in GPT-2 Small that causally control the GP task, show that scalar steering produces dose-response curves, and demonstrate 100% flip under $\sigma \times 2.0$ amplification.
2. **From 1D directions to 2D compasses (Section 4).** We formalize the semantic compass as a pair of OV singular vectors and define the joint amplitude-linearity + phase-stability test that separates causal compasses from geometric accidents.
3. **Four-model gender compass (Section 5).** Every compass produces $R^2 \geq 0.993$ amplitude linearity and $\Delta\phi \leq 3^\circ$ phase stability across $\alpha \in \{1, 3, 10\}$ and across four models spanning two orders of magnitude in parameters.
4. **Blind causal scan (Section 6).** A scan of all top-4 SV-pairs per head rediscovered the same sites without decoding, with head-level concentration significance $p < 10^{-6}$, and surfaces heads the decode survey missed.
5. **Non-pronoun compasses in Phi-3 (Section 7).** The recipe generalizes off the gender axis: a temporal-unit dial and an entity-type dial in Phi-3, both with linear α -scaling and stable phase.
6. **Rank-2 downstream transfer and asymmetry (Section 8).** The Stage-1 GP-to-occupation transfer and the Stage-1 masculine-vs-feminine asymmetry both persist under rank-2 compass intervention, confirming that the compass is not only a diagnostic but a steering primitive with the same behavioral signatures as its 1D projection.

2. Related Work

Circuit discovery. Circuit analysis reverse-engineers behavior into subsets of heads and MLPs (Olah et al., 2020; Elhage et al., 2021; Wang et al., 2023; Conmy et al., 2023; Nanda et al., 2023; Meng et al., 2022; Marks et al., 2024). Head-level ablation and patching are the standard tools. We work one level finer: inside a head, where the OV map’s singular structure is the interface between the head’s “read-in” and “write-out” behavior.

Linear feature discovery. The linear representation hypothesis (Park et al., 2024), DAS (Geiger et al., 2024), geometry-of-truth probes (Marks & Tegmark, 2024), sparse autoencoders (Cunningham et al., 2023; Templeton et al., 2024), and function vectors (Todd et al., 2024) all search for low-dimensional causal features. These methods search in *activation space*; our compasses are read directly from the *weight matrix*, are localized to a specific attention head, and give a continuous family of θ -rotations rather than a single direction. Helical / circular numerical features (Engels et al., 2024) are the closest geometric precedent, though they are estimated via probes on cached activations rather than read off the OV weights.

Activation steering. Activation addition (Turner et al., 2023) and inference-time intervention (Li et al., 2023) add a fixed vector to the residual stream to steer generation. Compass-based steering replaces the fixed vector with a continuous rank-2 dial: each rotation angle in $[0, 2\pi)$ is a valid intervention, and the amplitude-phase decomposition gives a principled way to pick one.

3. Sub-Component Units: Directions as Causal Axes

This section establishes the Stage-1 claim: individual OV singular directions causally control GP behavior in GPT-2 Small.

3.1. Circuit discovery by learnable masking

For each attention head at layer ℓ we compute two SVDs: the effective OV map $W_{OV}^{(\ell, h)} = W_O^{(\ell, h)} W_V^{(\ell, h)} = U^{OV} \text{diag}(S^{OV}) (V^{OV})^\top$, which controls residual-stream writes, and the effective QK map $W_{QK}^{(\ell, h)} = U^{QK} \text{diag}(S^{QK}) (V^{QK})^\top$, which controls attention patterns. We attach a learnable soft mask $m_i \in [0, 1]$ to every singular value, substitute $S \leftarrow m \odot S$ during the forward pass, and train the mask with the composite loss

$$\mathcal{L} = \text{KL}(P_{\text{full}} \parallel P_{\text{masked}}) + \lambda_1 \|m\|_1, \quad (1)$$

where P_{full} is the output distribution of the unmasked model and P_{masked} uses the gated SVD. Training with KL target

Table 1. Stage-1: gender-sensitive OV singular directions in GPT-2 Small recovered by learnable masking.

Direction	Coding	Role
L9.H7.SV1	Masculine	Primary he-control
L11.H8.SV6	Masculine	Secondary he-control
L11.H8.SV9	Masculine	Tertiary he-control
L10.H9.SV0	Feminine	Primary she-anchor
L8.H3.SV8	Feminine	Auxiliary she-control

0.10, L1 target 3500, learning rate 0.02, and early stopping (patience 5) converges to a sparse configuration in which the OV masks identify a handful of gender-relevant directions (Figure 1).

Table 1 reports the five discovered directions. We refer to them by their $LX.HY.SVz$ signature, where X is the layer, Y is the head, and z is the singular-vector index.

3.2. Scalar causal steering

Given a discovered direction (u_i, σ_i, v_i) at layer ℓ , we read off its per-token scalar coefficient $\alpha_i^{(t)} = V'^{(t)} \cdot u_i$, where $V'^{(t)}$ is the pre-attention residual at token t . A scalar intervention $\alpha_i^{(t)} \rightarrow \alpha_i'^{(t)}$ perturbs the residual stream by $\Delta_{\text{resid}} = (\alpha_i'^{(t)} - \alpha_i^{(t)}) \sigma_i v_i$. We evaluate four steering modes: *mean swap* (replace with opposite-gender empirical mean), *additive* ($\alpha_i' = \alpha_i + \delta$), *sigma-scaled* ($\alpha_i' = \alpha_i + k\sigma_{\text{emp}}$), and a continuous *sweep* over k .

3.3. Behavioral effect on the GP task

Figure 2 summarizes the effect on the GP task (150 he prompts, 156 she prompts). On HE prompts, sweeping toward feminine values reduces accuracy from 84.00% $\rightarrow$ 32.67% and target logit-difference from 2.525 $\rightarrow$ 0.228. On SHE prompts, sweeping toward masculine values is markedly weaker: accuracy drops only to 53.21% (2.840 $\rightarrow$ 1.480 LD). Under $\sigma \times 2.0$ smart-ablation amplification on HE prompts, accuracy collapses to 0.00% with target LD -2.143 , i.e. a full 100% he $\rightarrow$ she flip. Direction-selective interventions confirm specificity: masculine-only swap on HE reduces accuracy to 45.33%; feminine-only swap on SHE reduces it to 57.05%.

The asymmetry—strong masculine control, weaker feminine control—is the single most consistent phenomenon across every Stage-1 experiment. We return to it in Section 8.

Summary. Five OV singular directions in GPT-2 Small causally control GP behavior; scalar steering along them is dose-responsive and can be amplified to full behavioral reversal. The rest of the paper asks what *geometry* these directions live in.

4. From Directions to Compasses

The Stage-1 directions include GPT-2 L9H7.SV₁ (masculine) and GPT-2 L10H9.SV₀ (feminine). Are these isolated 1D axes, or are they one column of a richer 2D structure? We now define the rank-2 test that decides this.

4.1. Compass plane

For attention head h at layer ℓ we compute the combined OV map $W_{OV}^{(\ell,h)} = W_V^{(\ell,h)} W_O^{(\ell,h)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and its SVD $U \Sigma V^\top$ with columns $u_1, \dots, u_{d_{\text{model}}}$ and singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$. A *compass plane* for the head is an ordered pair of columns $C_{i,j}^{(\ell,h)} = \text{span}(u_i, u_j)$, $i \neq j$. We always take left-singular columns, the *write-out* directions, because they correspond to what the head writes into the residual stream rather than what it reads.

4.2. Causal α -sweep

The compass test is a causal injection that varies an *angle* θ and a *scale* α :

$$v(\theta, \alpha) = \alpha \sigma_i \cos \theta u_i + \alpha \sigma_j \sin \theta u_j. \quad (2)$$

For a prompt p whose final-token residual stream just before layer ℓ is r , we hook at `blocks.{ $\ell$ }.hook_resid_pre` and replace $r \leftarrow r + v(\theta, \alpha)$; the remainder of the forward pass runs unmodified. We record

$$\text{LD}(\theta, \alpha; p) = \text{logit}_{t_+} - \text{logit}_{t_-} \quad (3)$$

for antipodal probe tokens (t_+, t_-) , average over three prompts per condition, sweep θ through 36 angles (step 10°), and run $\alpha \in \{1, 3, 10\}$.

For each curve we fit

$$\text{LD}(\theta) \approx \mu + A \cos(\theta - \phi) \quad (4)$$

by the first DFT bin and record mean μ , amplitude A , phase ϕ , and the angles of maximum / minimum response.

The joint test. A random plane produces neither amplitude nor phase structure. A plane that correlates with the category without being causally read produces some amplitude but no dependence on α . A truly causal compass satisfies two conditions simultaneously: *amplitude linearity* ($A(\alpha)$ grows linearly with α , $R^2 \geq 0.95$) and *phase stability* (ϕ drift $\Delta\phi \leq 10^\circ$ across $\alpha \in \{1, 3, 10\}$). We enforce both.

4.3. Nine-test battery

We complement Equation (2) with a battery of eight null and falsification tests: vocabulary decoding, 2π -cyclicity, label permutation, random-plane null, principal-angles check

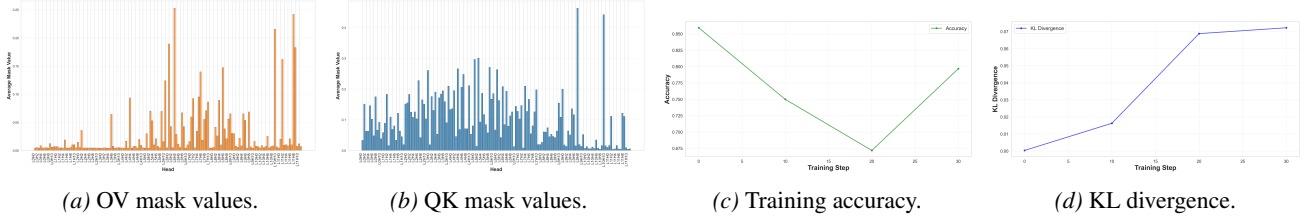


Figure 1. Circuit discovery by learnable masking over SVD directions. OV and QK mask averages (a,b) converge to sparse configurations; training accuracy (c) and KL divergence (d) stay controlled.

against unrelated heads, scan-planes (target pair is $\arg \max$ over $i, j \leq 10$), semantic ablation (zeroing P_{Cr} collapses category-conditioned logits), causal patching (swapping P_{Cr} between $\pm$ -context prompts flips the logit sign), and an end-to-end self-test. Full specifications are in Section A.

4.4. Models

We use TransformerLens (Nanda & Bloom, 2022) with HookedTransformer on bfloat16. Target heads:

- **GPT-2 Small** (Radford et al., 2019): L9H7 (u_1, u_2) (decode-discovered) and L10H9 (u_0, u_3) (scan-discovered). Stage-1 directions L9H7.SV₁ and L10H9.SV₀ are individual columns of these planes.
- **Phi-3 Mini** (Abdin et al., 2024): L24H10 (u_0, u_1); L28H1 (u_0, u_3); L24H17 (u_1, u_2) (temporal); L24H28 (u_3, u_7) (entity).
- **Gemma-2-2B** (Team, 2024): L21H4 (u_0, u_2).
- **Llama-3.2-3B** (Grattafiori & Team, 2024): L26H14 (u_2, u_3) (scan-discovered).

5. Four-Model Gender Compass

Table 2 summarizes the causal α -sweep for the gender-pronoun compass across four models. Every compass produces amplitude that grows near-linearly with α ($R^2 \geq 0.993$) and phase that is stable to $\pm 3^\circ$ across the three α settings. Figure 3 shows representative polar plots. All injections are *in-context* at the head’s own layer, so fitted amplitudes reflect causal propagation through every subsequent block, not readout proximity.

Bridge to Stage-1. The GPT-2 L9H7 row uses (u_1, u_2); Stage-1 isolated u_1 alone. The GPT-2 L10H9 row uses (u_0, u_3); Stage-1 isolated u_0 alone. In both cases the Stage-1 1D direction is one column of the Stage-2 2D compass plane, and the compass exposes additional structure: a continuous θ -family of interventions, of which the Stage-1 direction is a single point.

Decoding the compass basis. Projecting write directions through W_U reveals the vocabulary each pole of the plane maps to. At the context-favoring pole the compass basis decodes coherently: *his, He, he* vs. *herself, her, She* on GPT-2 L9H7; *he, him, his, ihn* plus Japanese and Chinese masculine pronouns vs. *she, her* plus Japanese, Chinese, and Korean feminine pronouns on Gemma-2-2B (the compass is cross-lingual); matching patterns on Phi-3 and Llama. Multiple languages’ pronouns co-localize in the same plane, consistent with a shared gender axis in the write-out subspace.

6. Blind Causal Scan

The compasses above were surfaced by decoding the top- k singular vectors through W_U and reading promoted vocabulary. That recipe is biased toward visually clean cases. To rule out selection bias, we complement decode-driven discovery with a blind causal scan that applies the two-pillar test of Table 2 to every (L, H, u_i, u_j) candidate without inspecting decode.

Setup. For each model we enumerate all (layer $\times$ head $\times$ $\binom{4}{2}$) planes from the top-4 SVs of each head’s W_{OV} : 864 planes for GPT-2, 1344 for Phi-3, 240 for Gemma, 4032 for Llama. We run the gender α -sweep in-context with $\alpha \in \{3, 10\}$ and 12 angles, fit a sinusoid, and pass a plane if $R^2 \geq 0.95$, $\Delta\phi \leq 10^\circ$, and slope $A(10)/10 \geq 0.2$. Two null baselines: a *top-4* null (random orthonormal pair from the head’s top-4 write subspace) and a *full-OV* null (random orthonormal pair from $\mathbb{R}^{d_{\text{model}}}$).

Every model exceeds its full-OV null by a non-overlapping margin. Against the top-4 null the rates overlap, showing that the top-4 write subspace is itself semantically enriched; the discriminating statistic is *head-level concentration*, not aggregate rate. On GPT-2, 15/16 passing planes come from three heads (L9H7, L10H9, L11H1); under the independence null implied by aggregate rate, the probability of a single head hosting ≥ 4 passes is $\binom{6}{4}(0.02)^4 \approx 10^{-6}$. On Llama, 7/9 passes concentrate at two heads (L26H14 with 6/6 passes, L20H13 with 3/6). Phi-3 concentrates at L24 and L28.

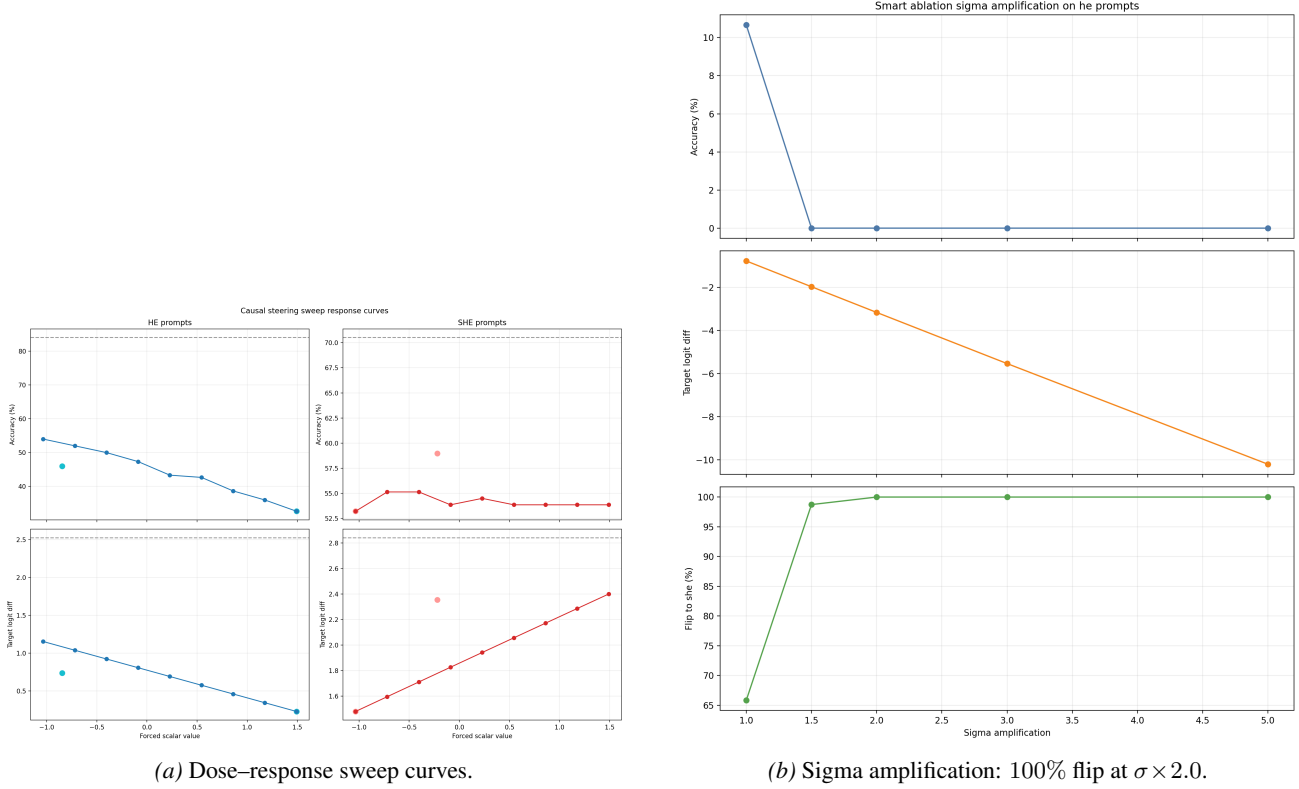


Figure 2. Stage-1 scalar causal steering on GPT-2 Small, GP task. Left: sweep produces dose-response curves; HE prompts respond more strongly than SHE prompts (asymmetric control). Right: σ amplification achieves a complete 100% he $\rightarrow$ she flip, ruling out confounding explanations of the steering signal.

Strengthened Monte-Carlo nulls. On Phi-3 L28H1 we run a 1000-trial random-plane null (observed = 40.77, random-mean = 20.47, $p = 0.142$; long-tailed because random directions occasionally land on other semantic axes) and a 2000-trial label-permutation null (perm-mean = 9.42, perm-max = 24.35, $p = 0.000$). Permutation is the sharp test.

The scan found heads the decode survey missed. GPT-2 L10H9, Phi-3 L28H1, and Llama-3.2-3B L26H14 were not in the original decode-based pick. L10H9’s gender content is concentrated on SV_0 alone (the decode browser preferred L9H7 because it spreads content across two orthogonal SVs); the scan ranks L10H9 first by amplitude slope (1.42 vs. L9H7’s 1.25). Decode browsing and causal scanning therefore do not pick up the same heads; a complete survey should use both.

7. Non-Pronoun Compasses in Phi-3

The same recipe surfaces two non-pronoun compasses in Phi-3: a *temporal-unit* dial (L24H17, (u_1, u_2) , $t_{\pm} = \text{month/hour}$) and an *entity-type* dial (L24H28, (u_3, u_7) , $t_{\pm} = \text{item/operation}$). Table 4 gives amplitudes and

phases; both satisfy amplitude-linearity and phase-stability at $\Delta\phi \leq 0.9^\circ$.

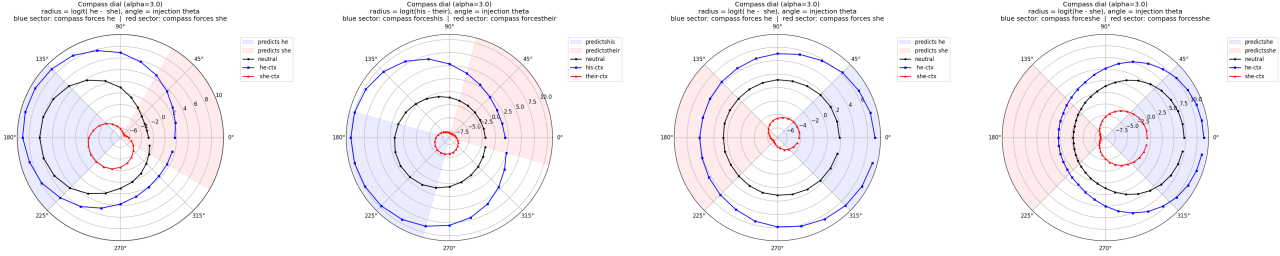
A two-harmonic fit separates *two-way* antipodal dials from *four-way* rotation dials. Using a four-token probe at $0^\circ/90^\circ/180^\circ/270^\circ$ and fitting $f(\theta) = \mu + A_1 \cos(\theta - \phi_1) + A_2 \cos(2\theta - \phi_2)$, the temporal compass L24H17 scores $A_2/A_1 = 0.32$ at $\alpha = 10$ (a genuine four-way dial: month / upon / hour / year), while the entity compass L24H28 scores $A_2/A_1 \leq 0.06$ (two independent antipodal axes). Decoding alone cannot distinguish these cases; the second harmonic does.

8. Rank-2 Downstream: Transfer and Asymmetry

A compass is useful only if a rank-2 rotation changes downstream behavior without destroying fluency. We test this on the same occupation-bias setup used in Stage-1 for scalar steering, now with the full rank-2 compass acting as the steering primitive. We use the GPT-2 L10H9 (u_0, u_3) compass, rotated toward its feminine pole (the θ at which the fitted $LD(\theta)$ is minimized on GP she-context prompts).

Table 2. Gender-compass causal sweep. $LD_{\pm}$: baseline $\text{logit}(t_+ - t_-)$ on $\pm$ -context prompts. $A(\alpha)$: fitted sinusoidal amplitude. $\Delta\phi$: phase drift across $\alpha \in \{1, 3, 10\}$.

Model	Head	(u_i, u_j)	(σ_i, σ_j)	LD_+	LD_-	$A(1)/A(3)/A(10)$	$\phi \pm \Delta\phi$
GPT-2 Small	L9H7	(1, 2)	(8.87, 8.46)	+5.31	-4.31	1.67/4.60/12.16	$-178.0^\circ \pm 2.9$
GPT-2 Small	L10H9	(0, 3)	(9.15, 6.80)	+3.73	-3.31	1.78/5.10/14.16	$+4.7^\circ \pm 0.3$
Phi-3 Mini	L24H10	(0, 1)	(9.42, 7.70)	+7.08	-6.97	0.67/2.01/5.22	$-154.9^\circ \pm 2.0$
Phi-3 Mini	L28H1	(0, 3)	(13.67, 7.61)	+5.06	-5.21	1.81/5.15/12.85	$+1.0^\circ \pm 1.3$
Gemma-2-2B	L21H4	(0, 2)	(2.35, 1.98)	+5.38	-5.21	0.18/0.59/2.19	$+0.2^\circ \pm 1.3$
Llama-3.2-3B	L26H14	(2, 3)	(1.14, 0.97)	+5.11	-4.58	0.28/0.88/3.01	$+161.6^\circ \pm 2.4$



(a) GPT-2, L9H7.

(b) Phi-3, L24H10.

(c) Gemma-2-2B, L21H4.

(d) Phi-3, L28H1.

Figure 3. Gender-compass polar plots at $\alpha = 3$. Radius: $\text{logit}(he - she)$; angle: injection θ . Blue/red: masculine/feminine-context prompts; black: neutral. The he/she sectors anchor at the fitted $\arg \max / \arg \min$.

Table 3. Blind causal-scan pass rates with 95% Agresti–Coull CIs. *Compass*: top-4 SV-pairs per head. *top-4 null*: random pair from the same top-4 write subspace. *full-OV null*: random pair from all of $\mathbb{R}^{d_{\text{model}}}$.

Model (planes/null)	Compass	top-4 null	full-OV null
GPT-2 Small (864/432)	2.06% [1.12, 3.01]	2.51% [1.04, 3.97]	0.44% [0.00, 1.06]
Phi-3 Mini (1344/672)	1.18% [0.60, 1.76]	1.62% [0.67, 2.57]	0.28% [0.00, 0.69]
Gemma-2-2B (240/120)	2.02% [0.25, 3.78]	1.55% [0.00, 3.73]	1.55% [0.00, 3.73]
Llama-3.2-3B (4032/2016)	0.22% [0.07, 0.37]	0.15% [0.00, 0.32]	0.00% [0.00, 0.09]

Table 4. Non-pronoun compasses in Phi-3 Mini.

Axis	Head	(u_i, u_j)	(t_+, t_-)	LD_+/LD_-	$A(1)/A(3)/A(10)$	$\Delta\phi$
Temporal	L24H17	(1, 2)	(month, hour)	+8.68/-6.83	0.69/2.10/7.60	0.9°
Entity	L24H28	(3, 7)	(item, operation)	+5.55/-3.53	0.29/0.88/3.06	0.0°

8.1. GP $\rightarrow$ occupation transfer

We evaluate on 36 occupation prompts of the form “The [occupation] said that” spanning 12 masculine-coded (doctor, engineer, scientist, lawyer, CEO, architect, pilot, surgeon, professor, programmer, executive, mechanic), 12 feminine-coded (nurse, receptionist, babysitter, housekeeper, assistant, secretary, hairdresser, librarian, caretaker, maid, nanny, florist), and 12 neutral (teacher, writer, manager, student, artist, journalist, accountant, therapist, consultant, designer, analyst, photographer) professions. We measure the pronoun bias-score change and the flip rate from stereotypical to counter-stereotypical continuation. With the calibrated feminine operating point ($\alpha = 2.00$, see Section 8.2), the compass produces an overall transfer flip rate of 58.33%: 83.33% on masculine-coded occupations, 66.67% on neu-

tral occupations, 25.00% on feminine-coded occupations.

8.2. Per-direction calibration of the feminine pole

The Stage-1 asymmetry—masculine control stronger than feminine—survives the promotion to rank-2. The raw compass-at- $\alpha = 1$ setting passes only 19/150 GP flips on she-context prompts. We sweep the feminine-pole injection scale $\alpha \in \{0.25, 0.50, \dots, 2.30\}$ (Figure 7); performance improves monotonically through $\alpha = 1.25, 1.40, 1.55, 1.70$, breaking the 19/150 plateau at $\alpha = 1.70$ (20/150 flips) and reaching 21/150 at $\alpha = 2.15$ –2.30. Beyond $\alpha = 2.00$ the transfer improves but GP accuracy on original she-context prompts begins softening (from 36.67% at $\alpha = 1.70$ to 35.33% at $\alpha = 2.30$). This defines two operating points: a *balanced* setting ($\alpha = 2.00$, strongest pre-bend configuration) and an *aggressive* setting ($\alpha = 2.30$, strongest validated transfer at visible GP cost).

Asymmetry is a representation-level phenomenon. Two observations tie the calibration back to the main story. (i) The asymmetry is already present in the uncalibrated Stage-1 scalar-steering setup (Section 3.3) and reproduces unchanged when the same head’s compass plane is swept. (ii) Calibration raises feminine-pole transfer (from the 19/150 plateau to 21/150 at $\alpha = 2.15$) but cannot bridge the gap to masculine-pole transfer. We interpret the residual gap as a representational property of GPT-2 Small’s gender subspace rather than a calibration artifact: the feminine pole is

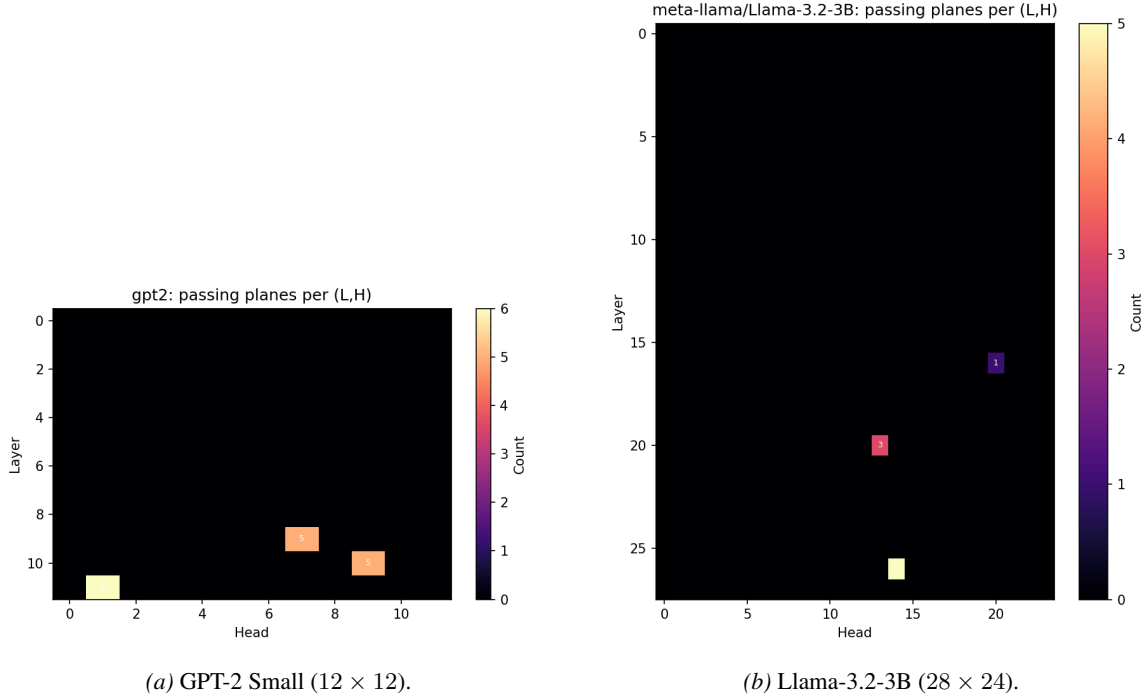


Figure 4. Blind causal-scan heatmaps: passing SV-pairs per (L, H) . Passes concentrate at a handful of heads in both models; full-OV null planes distribute uniformly and host zero such concentrations.

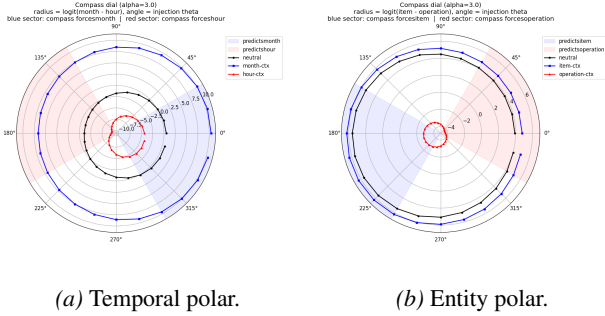


Figure 5. Non-pronoun Phi-3 compasses, $\alpha = 3$. Temporal (L24H17) rotates between *month* and *hour*; entity (L24H28) rotates between *item* and *operation*. Both show sinusoidal response with amplitude linear in α and stable phase.

simply written at smaller magnitude by the OV map than the masculine pole, and rotating inside the plane does not add write-strength the head never possessed.

9. Discussion

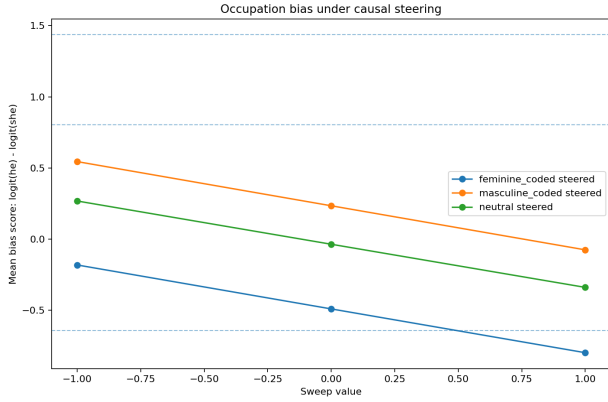
What a compass is. A semantic compass is a causal feature localized to an attention head: a 2D subspace of the head’s write-out space in which every angle is a valid intervention. The Stage-1 finding that individual OV singular directions are causal axes is therefore a 1D projection of

a richer structure; the compass makes the orthogonal axis explicit and turns a single-vector edit into a continuous θ -family.

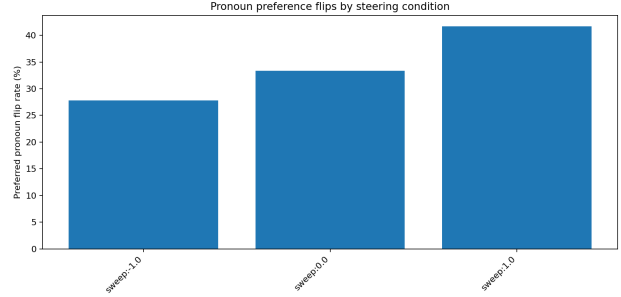
Why OV-SVD works as a discovery device. In a transformer block, W_{OV} determines the set of directions the head can add to the residual stream. Its top SVD columns are therefore the candidate “output languages” of the head. When a pair of such columns corresponds to a categorical axis, it must correspond to something the head *actually writes*, not just something we can correlate post hoc. This is the structural reason OV-SVD is a stronger discovery device than probing residual-stream activations directly.

Limitations. (i) *Prompt scope.* Baselines and injections average over three short ambiguous prompts per condition; robustness to paraphrase is not yet quantified. (ii) *Axis breadth.* We demonstrate gender on four models and temporal/entity on one (Phi-3). Sentiment, tense, formality, and animacy remain untested. (iii) *Asymmetry resolution.* Per-direction calibration narrows the masculine–feminine control gap but does not close it. Whether the gap is a fundamental property of how gender is stored in these models or can be mitigated by stronger calibration (or by combining multiple compass planes) is open.

Conclusion. Attention heads are not atomic: individual OV singular directions are causal control axes, and pairs of

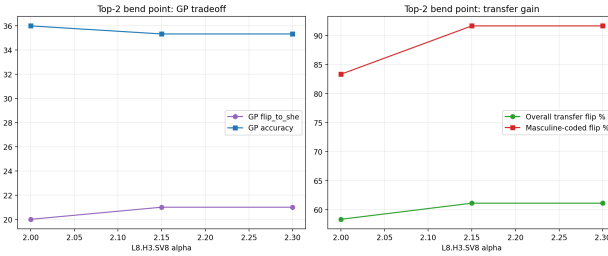


(a) Bias-score change across occupation types.



(b) Flip rates by occupation category.

Figure 6. Compass-driven GP→occupation transfer. Masculine-coded occupations respond most strongly to the feminine-pole rotation; feminine-coded occupations respond weakest. The rank-2 transfer behavior matches the asymmetric pattern seen at the 1D direction level in Section 3.



(a) Bend-point analysis: GP cost vs. transfer gain.

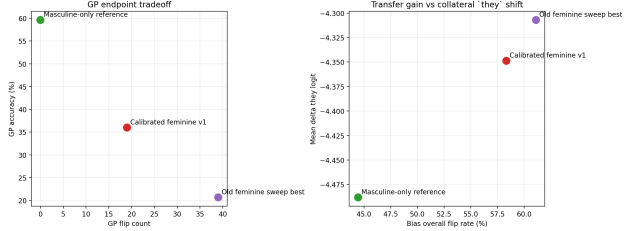

 (b) Endpoint trade-offs at balanced / aggressive α .

Figure 7. Feminine-pole calibration of the GPT-2 L10H9 (u_0, u_3) compass. Per-direction calibration narrows but does not eliminate the asymmetry first observed at the Stage-1 direction level; the masculine pole achieves 83.33% transfer on masculine-coded occupations, while the feminine pole reaches 25.00% on feminine-coded occupations at the balanced setting.

those directions form rank-2 compasses whose angular dial governs a categorical distinction. The same weight-space recipe uncovers gender dials in four decoder-only models and non-pronoun dials in Phi-3, recovers the same heads in a blind causal scan, and transfers the Stage-1 GP→occupation bias effect (including the masculine–feminine asymmetry) into a rank-2 steering primitive. OV-SVD is a probe-free, architecture-agnostic tool for discovering and editing low-dimensional causal features, with a continuous family of interventions rather than a single vector.

Impact Statement

This paper advances mechanistic interpretability by locating causal low-dimensional features to specific attention heads and showing that rotations within such a plane produce continuous, controllable changes in downstream behavior. The same rank-2 intervention that *reduces* stereotypical occupation continuations can, by rotating to the opposite pole, *inject* them: the primitive is dual-use. We restrict all downstream evaluations in this work to the Gender Pronoun task

and a public set of occupation prompts, and we do not release steering planes tuned to harm any demographic group. The persistent masculine–feminine control asymmetry we document is itself a finding about how current pretrained models represent gender and should be accounted for before compass-style interventions are used as a bias-mitigation tool in deployed systems.

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A. Nine-Test Battery

A compass must pass all nine of the following. Verdicts in parentheses are the *expected* outcomes (✓: signal preserved; ×: signal killed by the null or ablation, which is the desired falsification).

- **decode-heads** (✓): $W_U^\top u_i$ promotes readable categorical tokens at $0^\circ, 90^\circ, 180^\circ, 270^\circ$.
- **cyclicity** (✓): the polar angle of the compass projection wraps smoothly, 2π -periodic, without saturation.
- **permutation null** (×): shuffling the $\pm$ -context token set across prompts kills the compass signal.
- **random-plane null** (×): random orthonormal directions from $U(\text{SO}(d_{\text{model}}))$ remove the causal effect.
- **principal-angles** (✓): the fitted plane is not co-planar with planes from unrelated heads.
- **scan-planes** (✓): the target (i, j) is the arg max over (i, j) pairs with $i, j \leq 10$.
- **semantic-ablate** (×): zeroing P_{Cr} collapses category-conditioned logits toward neutral.
- **causal-patch** (✓): swapping P_{Cr} between $+$ -context and $-$ -context prompts flips the logit sign.
- **self-test** (✓): end-to-end pipeline reproducibility on a fixed prompt.

All four models pass all nine tests for the gender compass; Phi-3 additionally passes for the temporal and entity compasses.

B. Prompt Lists

All causal sweeps average over three prompts per condition. Prompts are short and semantically ambiguous to minimize baseline bias.

Gender (GPT-2, Phi-3, Gemma, Llama). *Neutral*: “The person said that”, “Then they said that”, “Afterwards, the speaker said that”. *Masculine-context*: “The man laced up his boots because...” (and analogous variants). *Feminine-context*: “The woman laced up her boots because...” (and analogous variants).

Temporal (Phi-3 L24H17, tokens *month/hour*). *Neutral*: “The event took about one”, “They waited nearly one”, “The project finished within one”. *Plus (month-biasing)*: “The rent is due every”, “Pregnancy lasts about nine”, “The magazine arrives once a”. *Minus (hour-biasing)*: “The meeting lasts exactly one”, “The drive takes roughly one”, “She slept for a single”.

Entity (Phi-3 L24H28, tokens *item/operation*). *Neutral*: “The list contained a single”, “He removed exactly one”, “The inventory showed one”. *Plus (item-biasing)*: “On the shelf sat a small”, “The box held only one”, “She picked up a single”. *Minus (operation-biasing)*: “The surgeon completed a complex”, “The factory ran one more”, “The team executed a risky”.

GP task (Stage-1, GPT-2 Small). 150 he-prompts and 156 she-prompts drawn from the Gender Pronoun coreference benchmark. Example: “The person who was sleeping in their car told the cop that”. The model must predict the target pronoun as the next token.

Occupation prompts (Stage-1 $\rightarrow$ Stage-2 transfer). 36 prompts of the form “The [occupation] said that”. *Masculine-coded* (12): doctor, engineer, scientist, lawyer, CEO, architect, pilot, surgeon, professor, programmer, executive, mechanic. *Feminine-coded* (12): nurse, receptionist, babysitter, housekeeper, assistant, secretary, hairdresser, librarian, caretaker, maid, nanny, florist. *Neutral* (12): teacher, writer, manager, student, artist, journalist, accountant, therapist, consultant, designer, analyst, photographer.

C. Learnable-Mask Training Hyperparameters

Stage-1 direction discovery (Section 3.1) on GPT-2 Small:

- Model: GPT-2 Small (124M parameters, 12 layers, 12 heads, $d_{\text{model}} = 768$).
- Batch size: 64. Epochs: 2.
- Learning rate: 0.02. Weight decay: 0.
- L1 weight: 1.95×10^{-4} .
- Temperature: 1.0. Evaluation interval: 50 steps. Validation interval: 20 steps.
- Early-stopping patience: 5 epochs. Minimum learning rate: 1×10^{-6} . Learning-rate reduction factor: 0.5.
- Target KL divergence: 0.10. Target L1 sparsity: 3500.
- Mask initialization value: 0.99. Sparsity threshold: 1.0×10^{-3} . Maximum iterations per batch: 15.

D. Stage-1 Full Results Table

Table 5. Complete scalar-steering results on the GP task (GPT-2 Small). *Sweep*: continuous scalar steering. *Smart all*: mean-swap on all five directions. *Smart masc only*: mean-swap on masculine directions. *Smart fem only*: mean-swap on feminine directions. *Sigma* $\times 2.0$: σ -amplified smart ablation.

Condition	Gender	Acc (%)	Target LD	they logit	Flips
Baseline	HE	84.00	+2.525	11.691	0
Baseline	SHE	70.51	+2.840	11.886	0
Sweep strongest	HE	32.67	+0.228	7.672	48
Sweep strongest	SHE	53.21	+1.480	7.256	12
Smart all	HE	10.67	-0.768	7.300	52
Smart all	SHE	46.79	+1.487	7.503	4
Smart masc only	HE	45.33	+0.770	7.222	24
Smart fem only	SHE	57.05	+2.232	7.423	2
Sigma $\times 2.0$	HE	0.00	-2.143	5.891	75

E. Feminine-Pole Calibration Sweep

Table 6. Per-direction calibration results for the feminine-pole rotation of the GPT-2 L10H9 (u_0, u_3) compass (Section 8.2). *GP flips*: she $\rightarrow$ he flips on the 150 GP she-context prompts. *Transfer*: flip rate on the 36 occupation prompts. *GP acc*: accuracy on the original GP she-context prompts.

α	GP flips	Transfer (%)	GP acc (%)	they logit
0.25	17	52.78	37.33	7.39
0.50	18	55.56	36.67	7.35
0.75	18	55.56	36.67	7.31
1.00	19	55.56	36.67	7.27
1.25	19	58.33	36.67	7.21
1.40	19	58.33	36.67	7.18
1.55	19	58.33	36.67	7.14
1.70	20	58.33	36.00	7.10
2.00	20	58.33	36.00	7.02
2.15	21	61.11	35.33	6.98
2.30	21	61.11	35.33	6.94

The plateau breaks at $\alpha = 1.70$ (20/150 GP flips) and again at $\alpha = 2.15$ (21/150). The bend point at $\alpha = 2.00$ defines the *balanced* operating point used for the transfer numbers in Section 8.1; beyond it, GP accuracy on original she-context prompts begins softening, marking the *aggressive* regime ($\alpha = 2.30$).